

SHOAIB IKRAM

AI/Machine Learning Engineer

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PROFESSIONAL SUMMARY

Passionate about blending technologies and turning ideas into impactful solutions for real-world problems. Focused on Artificial Intelligence, machine learning, deep learning, computer vision, natural language processing, model training, deployment, databases, and real-world AI applications.

EDUCATION

Bachelor of Science in Artificial Intelligence

Oct 2022 – Oct 2026

National University of Technology (NUTECH), Islamabad

F.Sc, Pre-Engineering (Grade A)

2020 – 2021

Fazaia Inter College Islamabad, E-9

Matriculation, Biology (Grade A)

2018 – 2019

Fazaia Inter College Islamabad, E-9

TECHNICAL SKILLS

Machine Learning: Model Training, Regression, Classification, Feature Engineering

Deep Learning: Neural Networks, CNN, Autoencoders | Computer Vision: Image Processing, Object Detection, YOLO

Natural Language Processing: NLP, LLM-related workflows | Programming: Python, SQL, JavaScript

Backend and APIs: FastAPI, REST APIs | Databases: PostgreSQL, PostGIS

Web Technologies: HTML, CSS, JavaScript, GeoJSON

Deployment and Tools: Model Deployment, Git, GitHub, VS Code | Simulation: Gazebo, PX4 SITL, QGroundControl, MAVLink

INTERNSHIP

AI/ML Intern — National Disaster Management Authority (NDMA)

PakEcoVision — Pakistan Atmospheric and Environmental Intelligence Portal

- Developed an interactive environmental intelligence portal for Pakistan covering population, CO2, oxygen, deforestation, forest coverage, and plantation-required statistics at province and district levels.
- Built a PostgreSQL/PostGIS database and FastAPI backend and integrated them with a dark-themed HTML, CSS, and JavaScript frontend.
- Implemented GeoJSON-based interactive mapping, API-driven province and district pop-ups, environmental scoring, status classification, forest statistics, and plantation requirements.
- Prepared an internship report documenting the project and proposed AI/ML enhancements using APIs such as Google Gemini, NASA Earthdata, and NASA FIRMS.
- Technologies: Python, FastAPI, PostgreSQL, PostGIS, HTML, CSS, JavaScript, GeoJSON, REST APIs.

PROJECTS

Final Year Project: Offensive Cyber Exploitation Of the UAV Communication Systems

- Studied UAV communication systems and cybersecurity vulnerabilities in the MAVLink protocol.
- Trained an unsupervised 1D Convolutional Autoencoder (TensorFlow/Keras) on normal flight telemetry — no attack samples seen during training.
- Achieved 100% detection rate, 0% false positive rate, and sub-2-second latency on the test set.
- Developed a Gazebo and PX4 SITL simulation environment integrated with QGroundControl.
- Carried out the project for offensive cyber attack purposes against UAVs and implemented two simulated attacks: GPS spoofing and C2 (Command and Control) channel compromise.
- Technologies: Python, PyMAVLink, MAVSDK, Wireshark, ROS 2, Gazebo, PX4 SITL.

COURSES AND CERTIFICATIONS

Agentic AI — Coursera (In Progress) | Artificial Intelligence — Coursera | Machine Learning — Coursera | Deep Learning and Neural Networks

Computer Vision and Image Processing | Natural Language Processing | Convolutional Neural Networks | Model Deployment

SQL and Database Management | Data Structures and Algorithms

AREAS OF INTEREST

Machine Learning | Deep Learning and Neural Networks | Computer Vision | Natural Language Processing | Model Training and Deployment

HOBBIES AND INTERESTS

Hiking | Horse Riding | Cricket | Badminton

CORE COMPETENCIES

AI Model Development | Data Analysis | Database Integration | REST API Integration | Geospatial Data | Environmental Data
Visualization | Full-Stack AI Applications | Cybersecurity Research

REFERENCES

References will be available on request.